

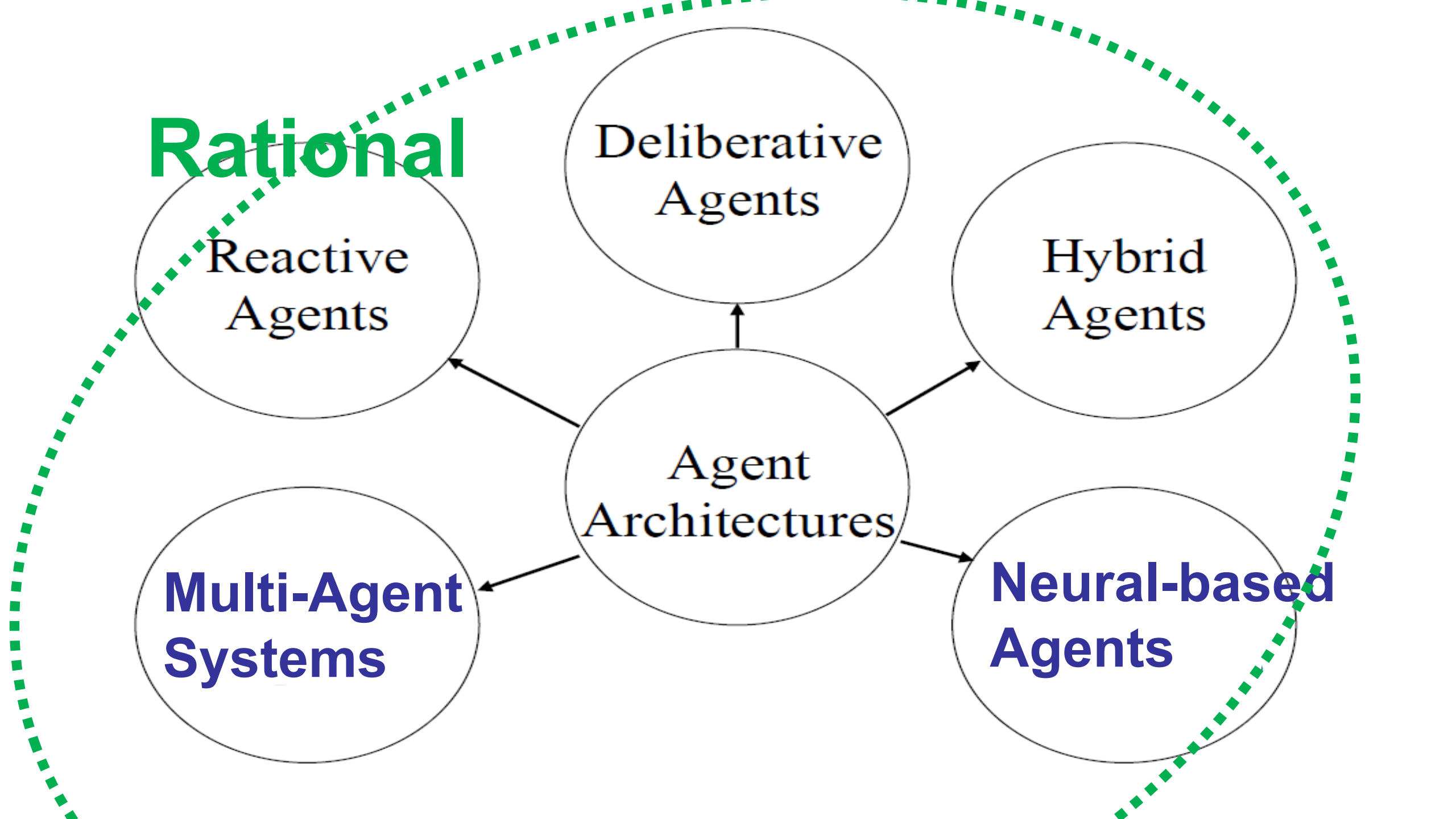
UNIVERSITY OF CAPE TOWN
Department of Computer Science

CSC3044S
Artificial Intelligence:
Agent Architectures III

Geoff Nitschke
gnitschke@cs.uct.ac.za

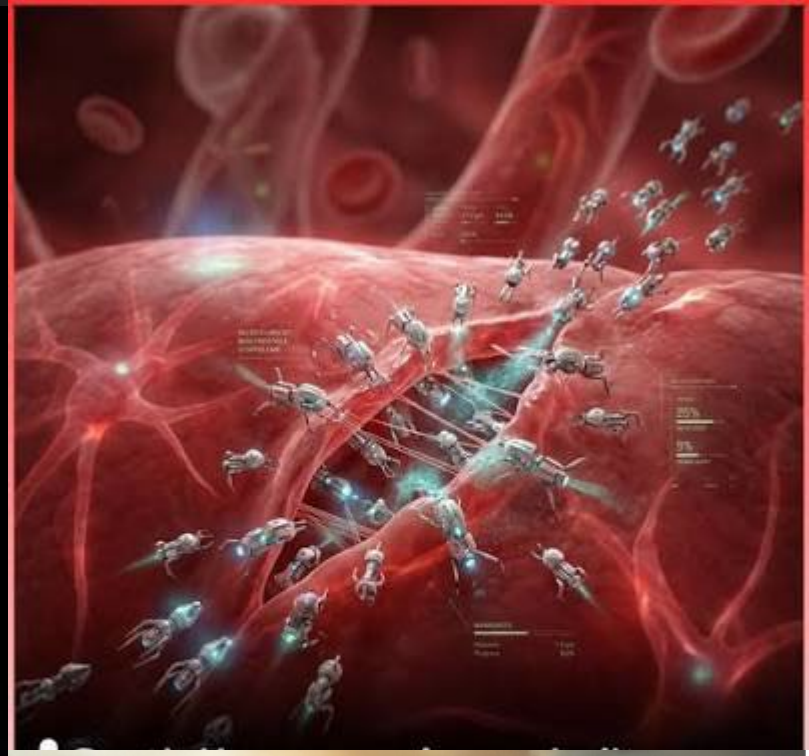
Agent Architecture Examples

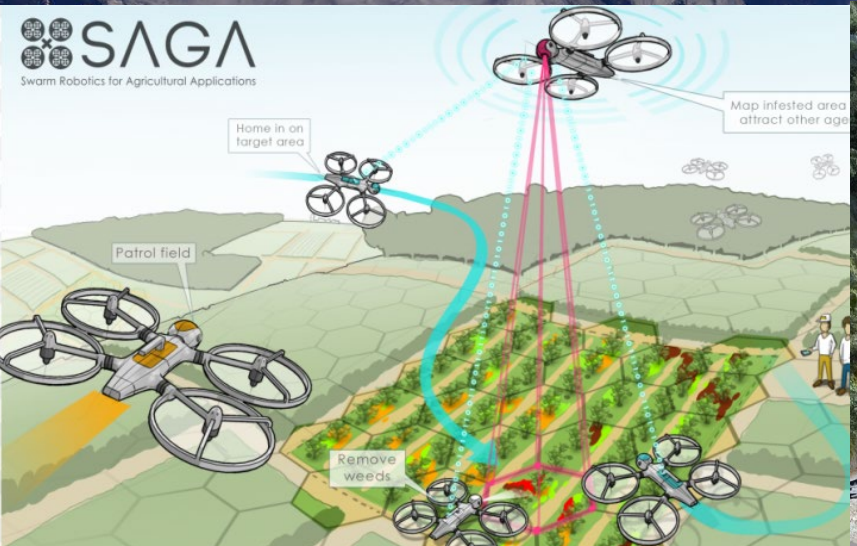
- **Reactive:** *Subsumption, Boids, Continuum, Stigmergy.*
- **Deliberative (Symbolic-based):** *Belief, Desires Intention (BDI).*
- **Cognitive (Neural-based):** AASMO, Soar, ACT-R, Brahms, LIDA, PreAct, Cougaar, PRODIGY, FORR, ….
- **Hybrid:** 3T, AuRA, Brahms, GAIuS, GRL, ICARUS, InteRRaP, TinyCog, *TouringMachines,* Claude, Gemini, DeepSeek, GPT…

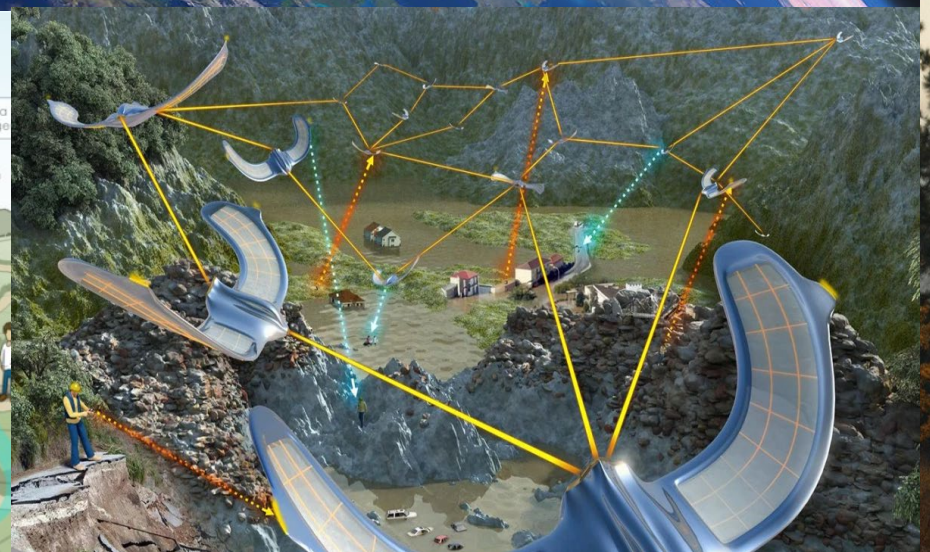

Rational
Deliberative Agents
Reactive Agents
Hybrid Agents
Agent Architectures
Multi-Agent Systems
Neural-based Agents

MAS Applications









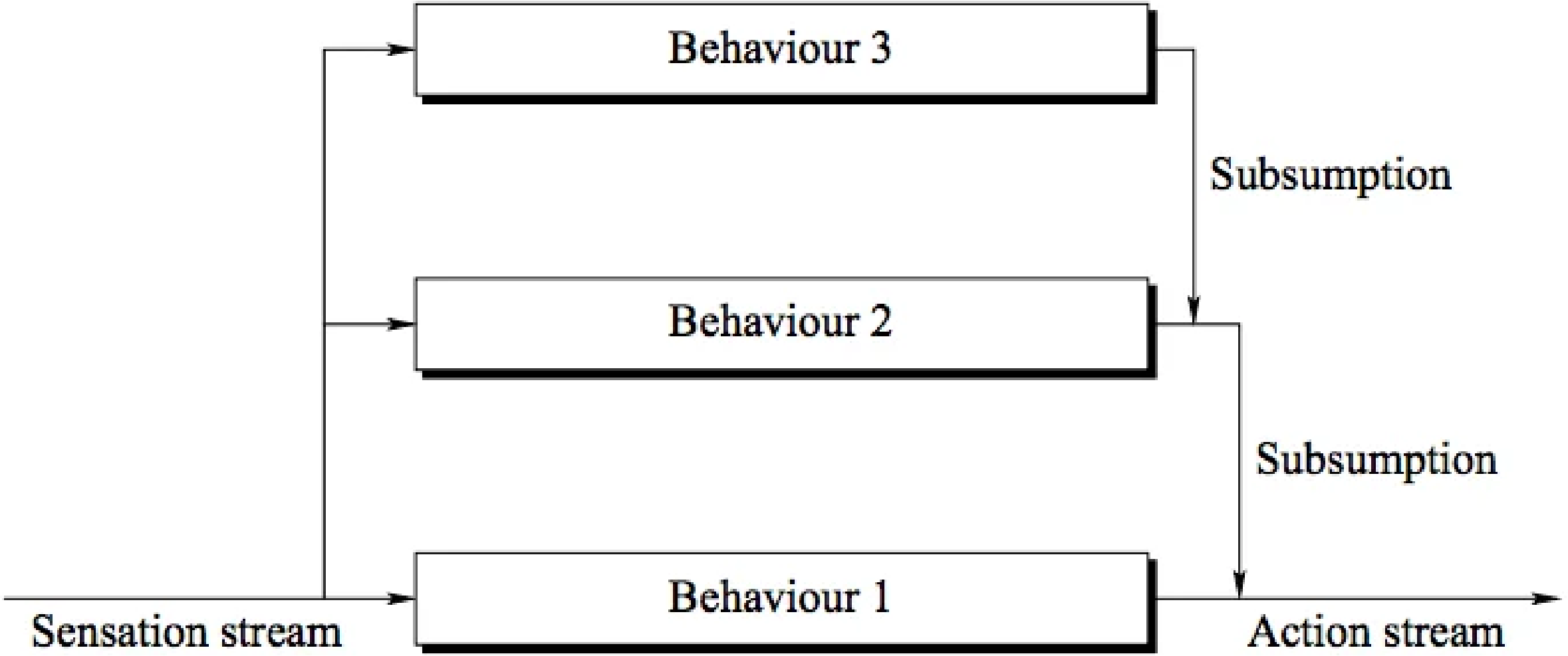


Multi-Agent Systems (MAS) and the Subsumption Architecture

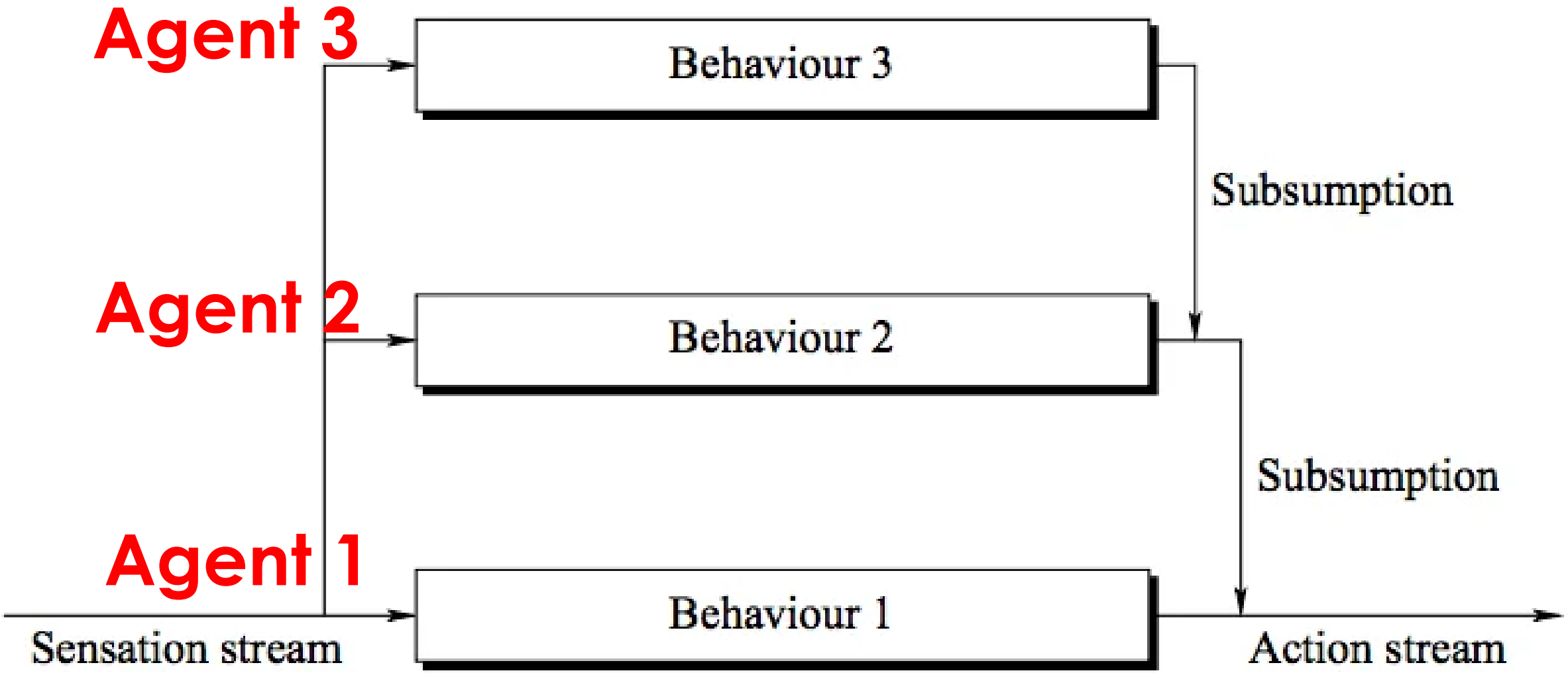
Designing Multi-Agent Systems Behaviour

- Multi-agent subsumption architecture:
  - 1st: Agents learn individual behaviours.
  - 2nd: Bottom-up control.
  - 3rd: Individual agents interact – Team behaviours *emerge*.

MAS Subsumption Architecture



Potential Application?




FESTO



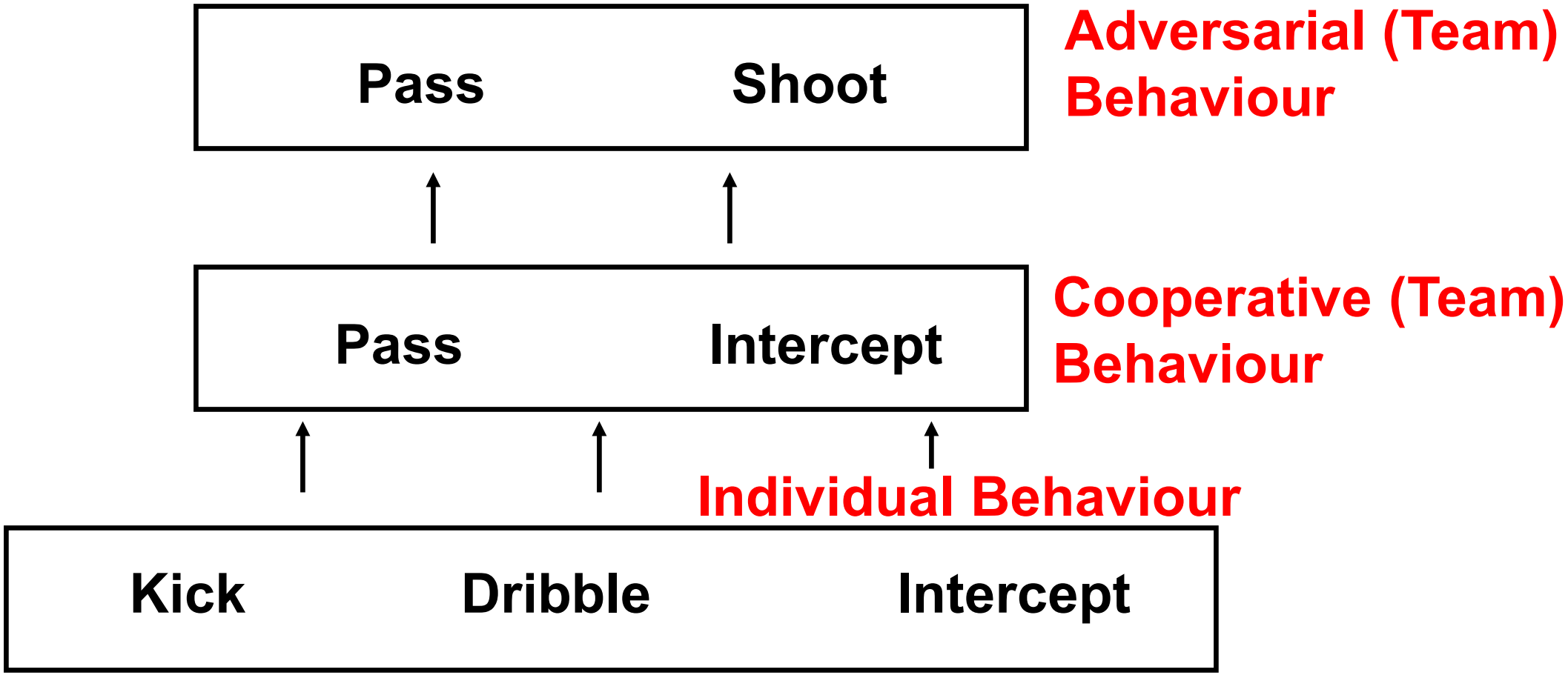

FiftyStorms3D 0:0 ODENS 11.0 2nd half
Playmode: PlayOn


kick off
connect
scale 1.0
detail 1
mode std
unzoom
quit
TeamA 0:0 TeamB
before_kick_off 65535

RoboCup: Designing Team Behaviour

Layer	Type	Behaviour	Example
1	Player-ball	Individual	Ball interception
2	1-to-1 player	Multi-agent	Pass or shoot?
3	1-to-N player	Team	Pass or shoot?
4	Team formation	Team	Strategic positioning
5	Team-to-Opponent	Adversarial	Strategic Adaptation

Game Playing Behaviours

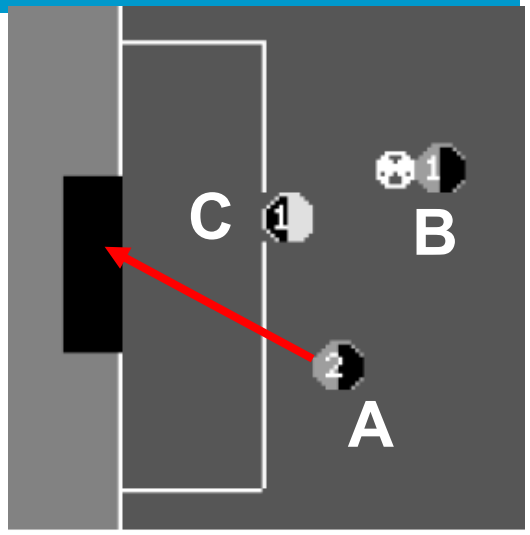


Game Playing Layers

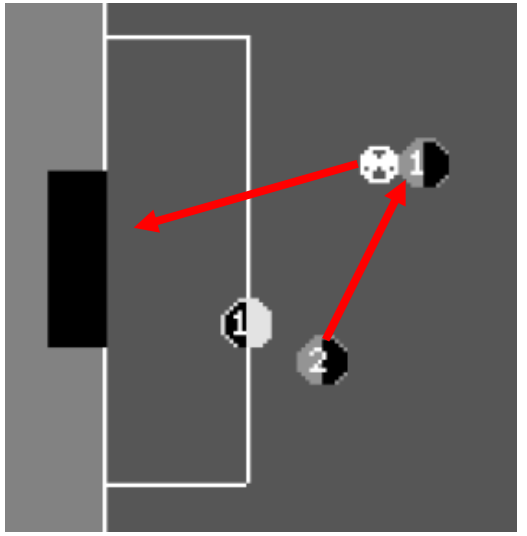
- **Individual:** Kick, Dribble (“hard-wired” rules).
- **Cooperative (2 agents):** Intercept, Pass.
- **Team (≥ 3 agents: team-mates, goalie):**
  - Pass, dribble or shoot?
- **Adversarial (2 teams of ≥ 3 agents):**
  - Pass, dribble or shoot?

Team Behaviour: Pass or Shoot?

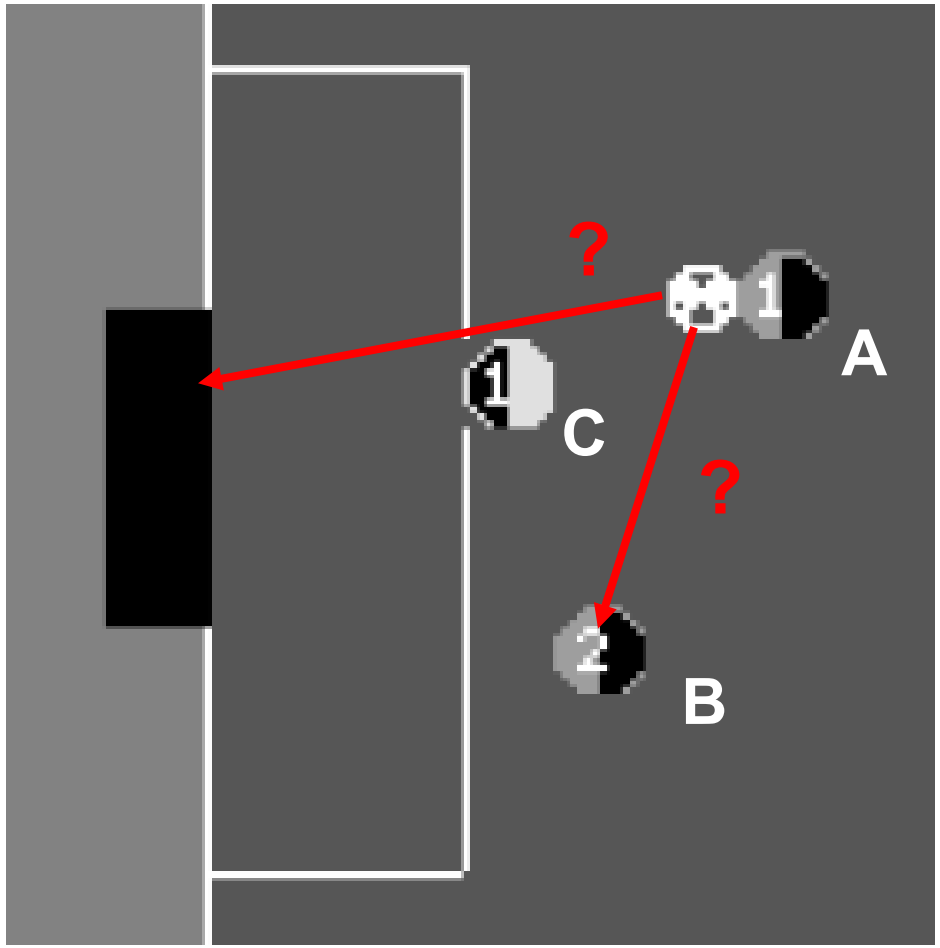
- **Agents A and B:**
  - Use learned shoot behaviour.
  - Must learn *pass or shoot* behaviour.



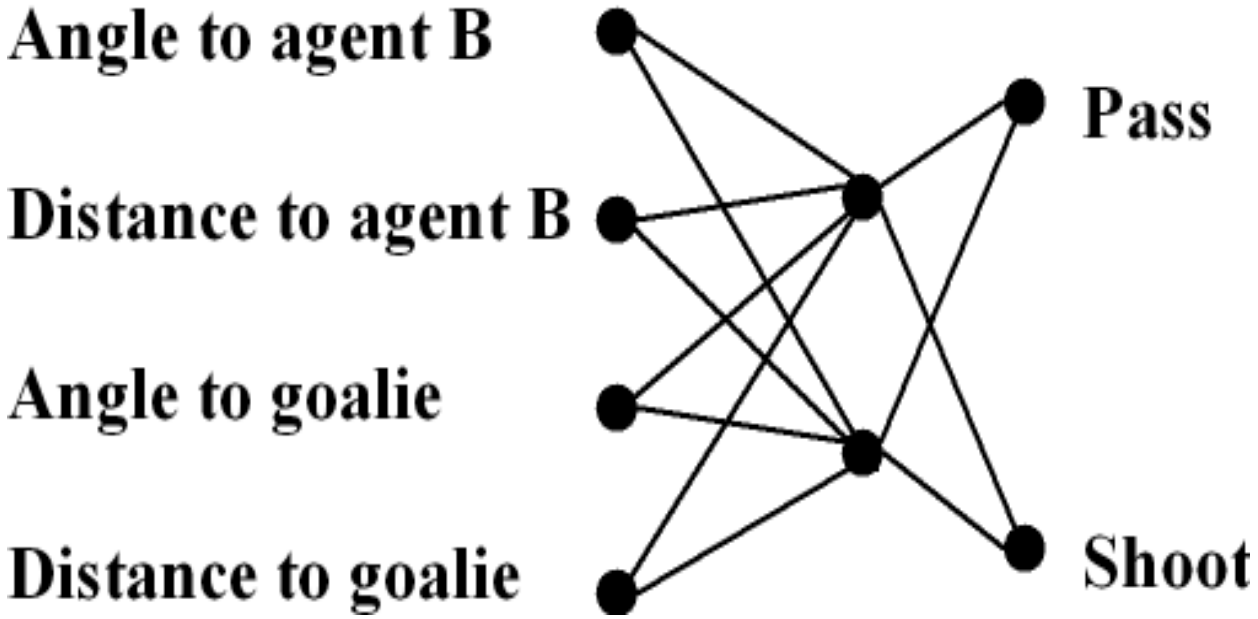
- **Agent C (Goalie):**
  - Uses learned intercept behaviour.

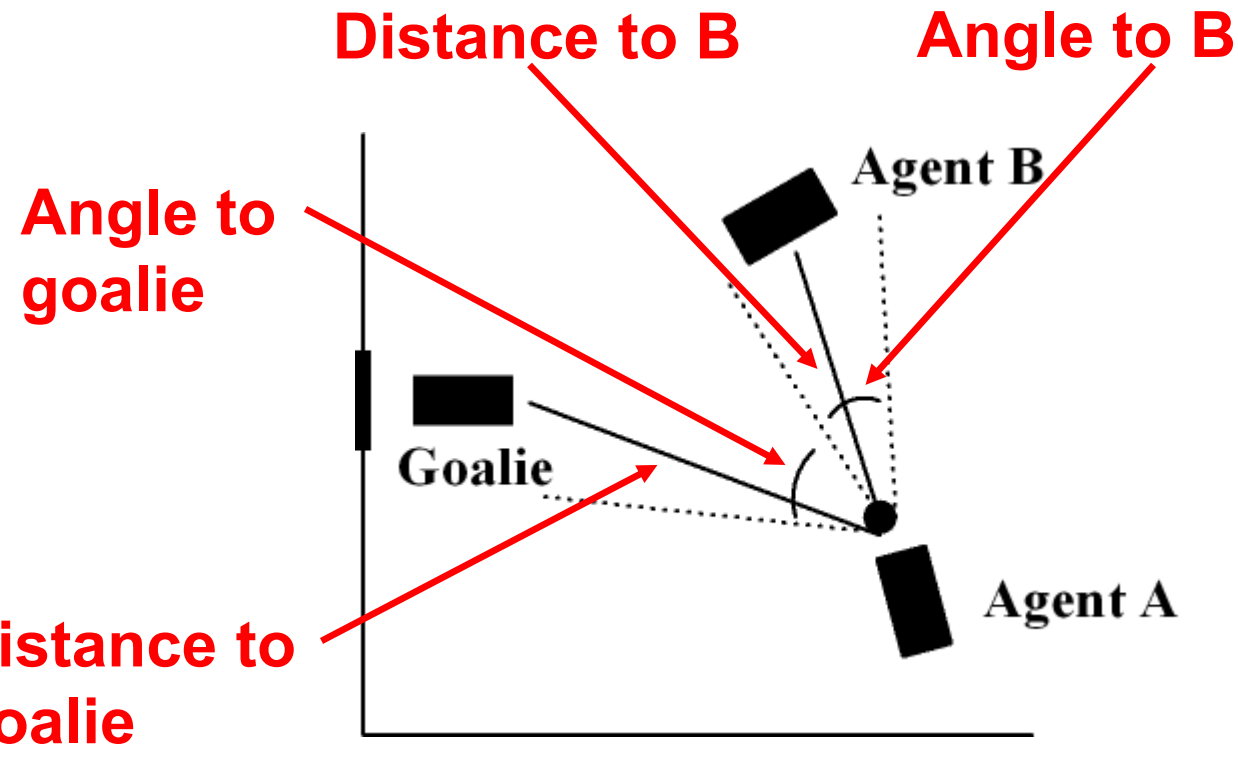


“Pass or Shoot” ANN Topology?



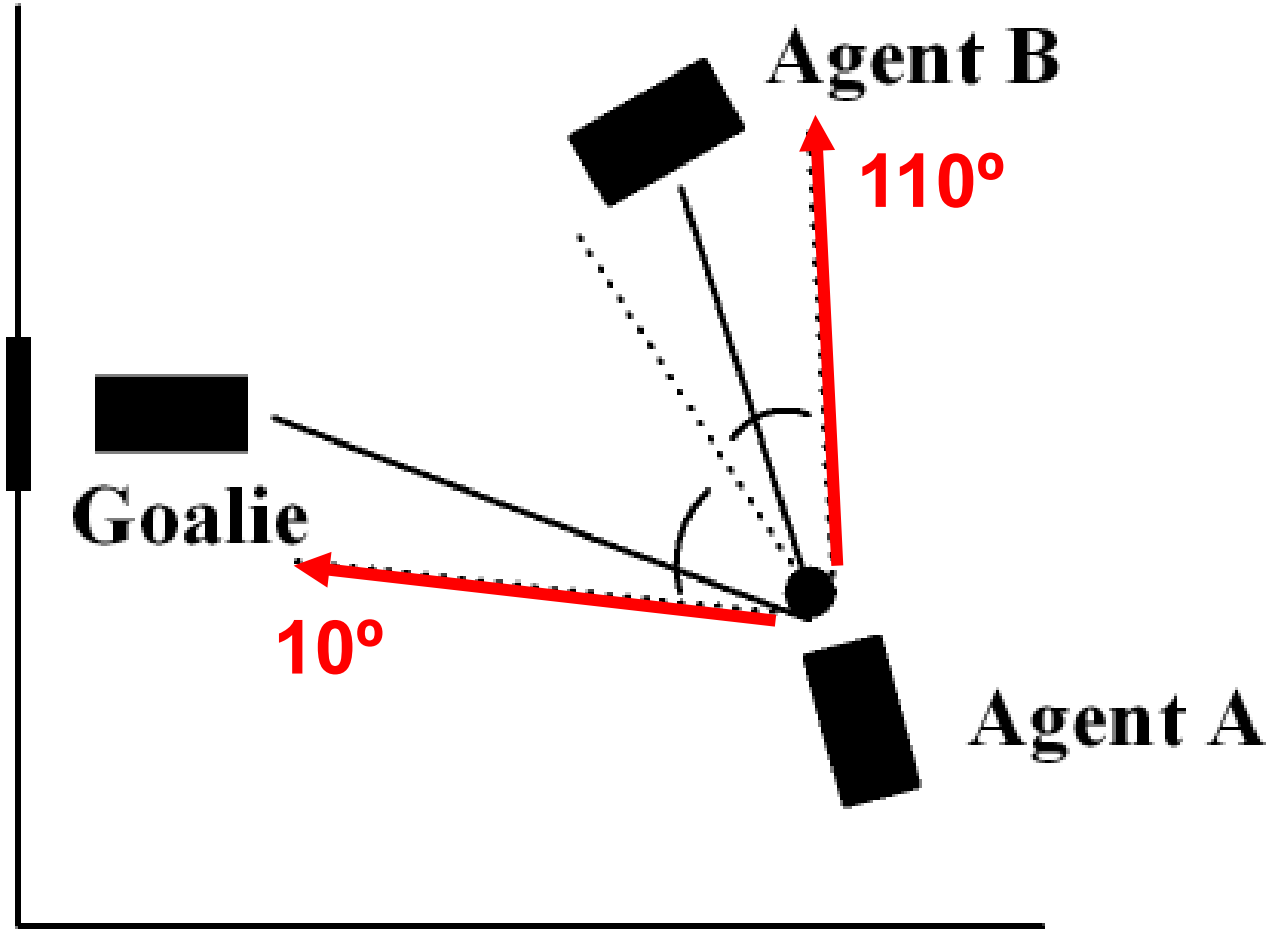
Agent A: Pass or Shoot ANN





What about Training Data?

Collecting Training Data



Training example:

{ ***Angle to B***, *Distance to B*, ***Angle to Goalie***, *Distance to Goalie* }

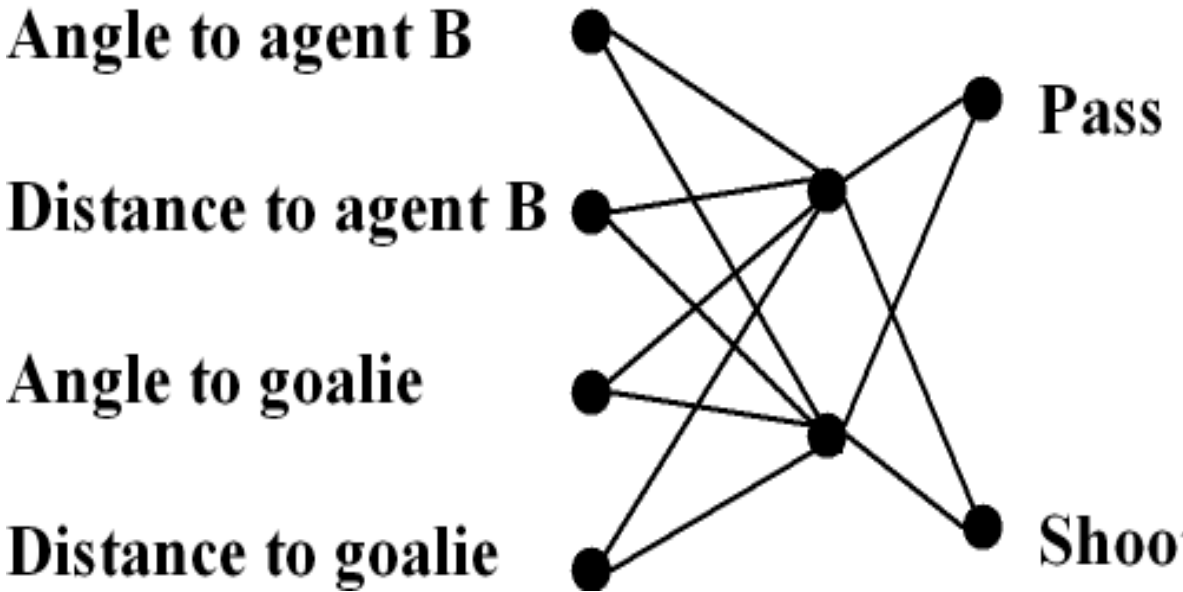
Generate Labelled Training Data

- **Run: 5000 simulations** (Agent A, B, Goalie start in random positions).
  - Agent A randomly *shoots* or *passes* (Agent B intercepts).
- **We get: 915 +*ve*** training examples **(1)**, **4085 –*ve*** examples **(0)**.
- **Labelled training example:** { *110*, *0.39*, *10*, *0.38, 0* }

  // { ° to B, DistToB(Normalised), ° to Goalie, DistToGoalie(Normalised), Label }
- **Goal scored →** Training example labeled “good” **(1)**
- **No goal →** Training example labeled “bad” **(0).**

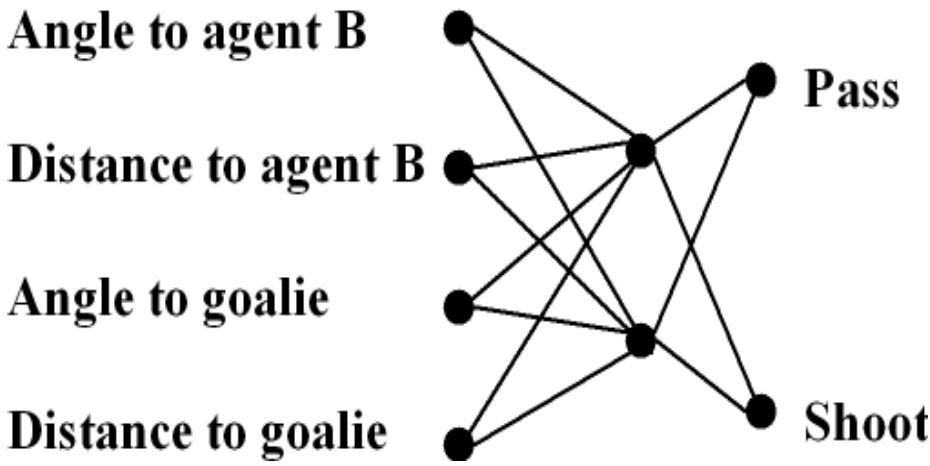
Agent A: "Pass or Shoot" Training

- Training parameters:
  - 5000 training examples.
  - Backpropagation weight adaptation.
  - Learning rate $n = 10^{-6}$
- Training results:
  - Mean squared error = 0.0598.

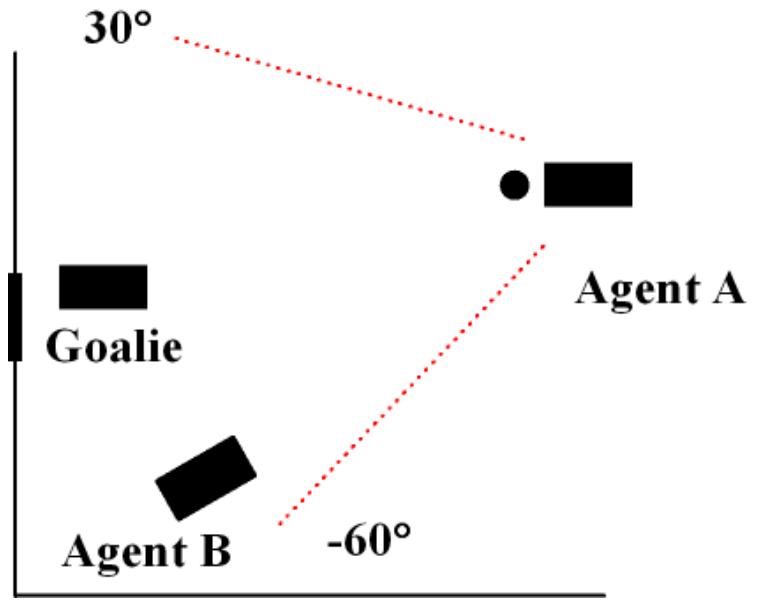


Testing Agent A: Pass or Shoot?

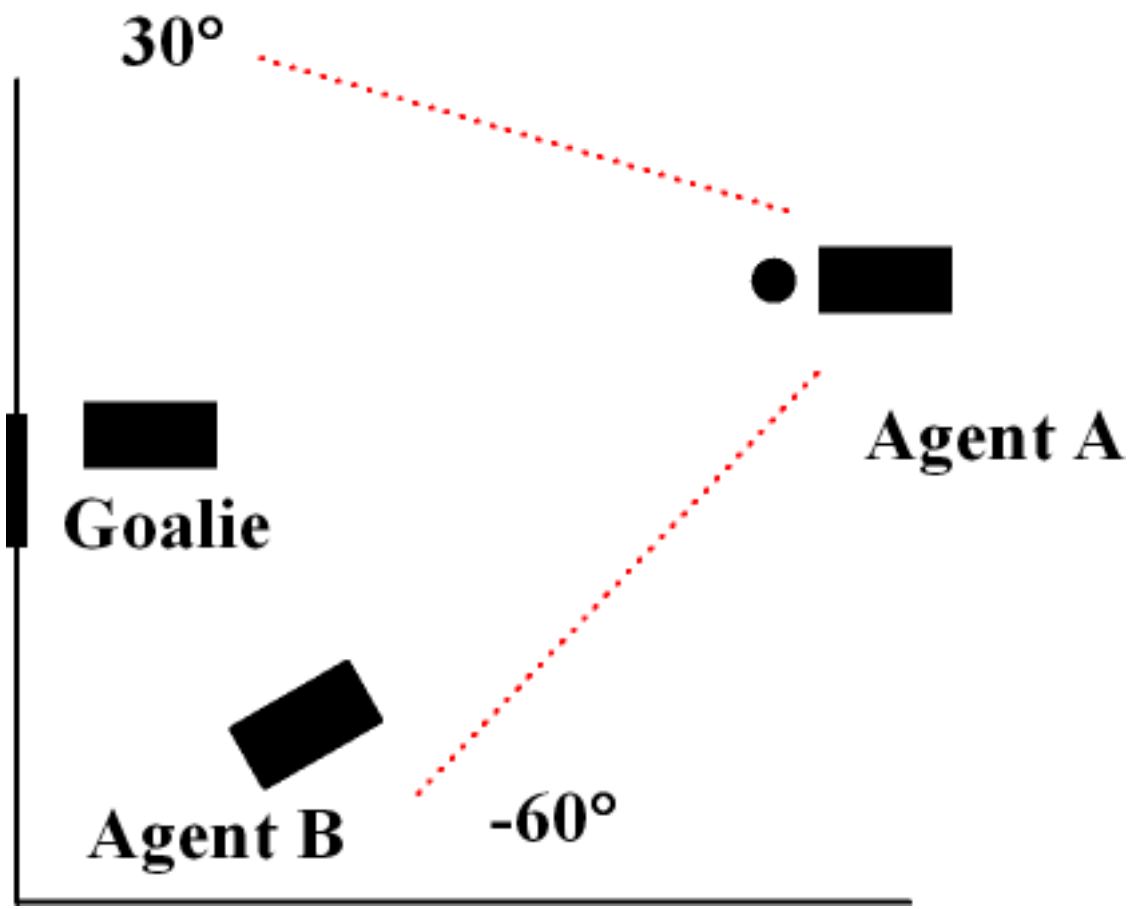
- **2000 Test simulations:**
  - Goalie, Agent B in random positions.
  - In angle range: **[-60º, 30º]** of **Agent A.**
  - Agent A's ANN receives 4 inputs: output?



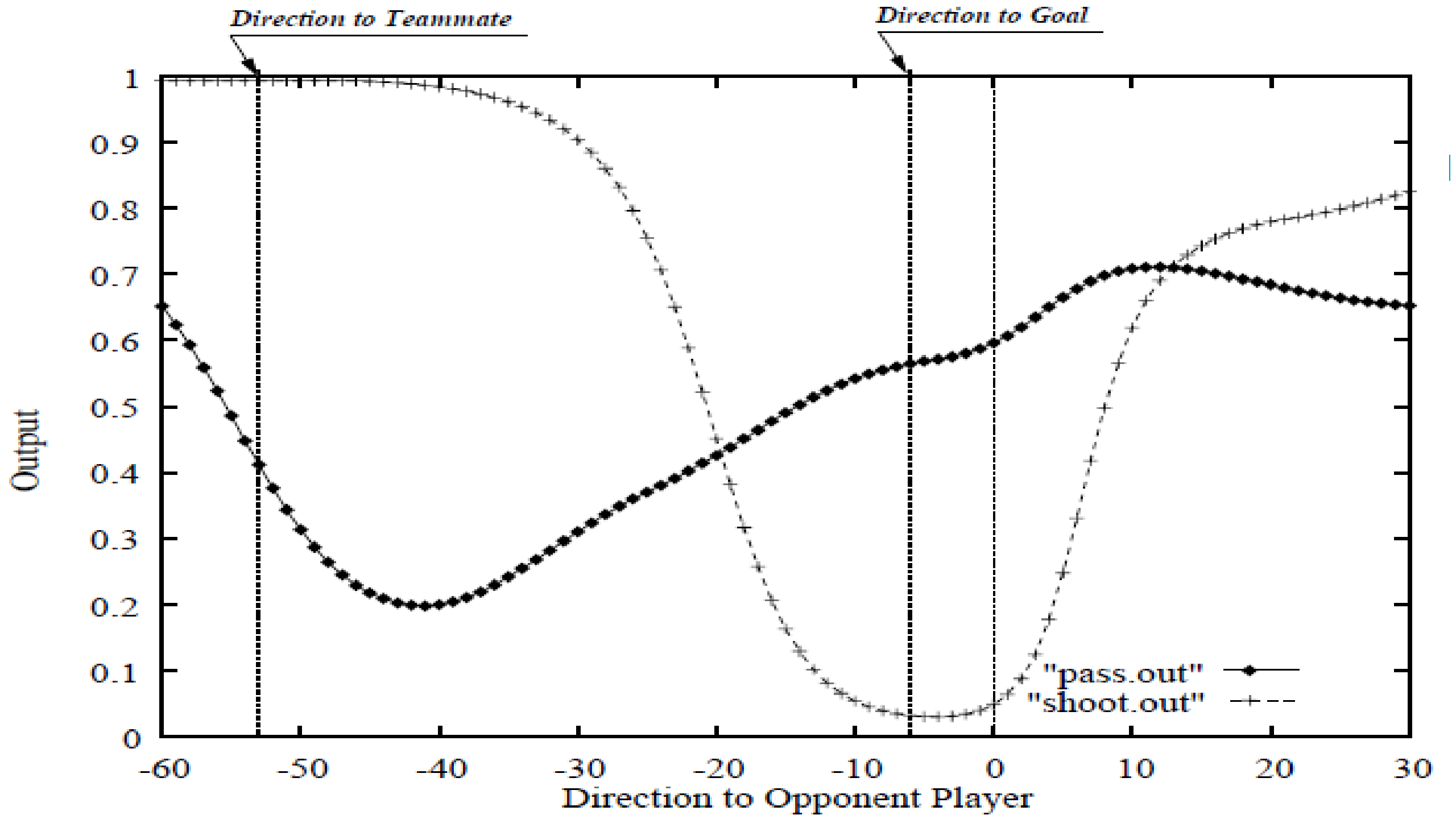
- **Agent A's output per tick:**
  - Node with highest activation → *Action selected*.
  - Output Node 1: *Pass*.
  - Output Node 2: *Shoot*.



Testing Agent A: Pass or Shoot ANN

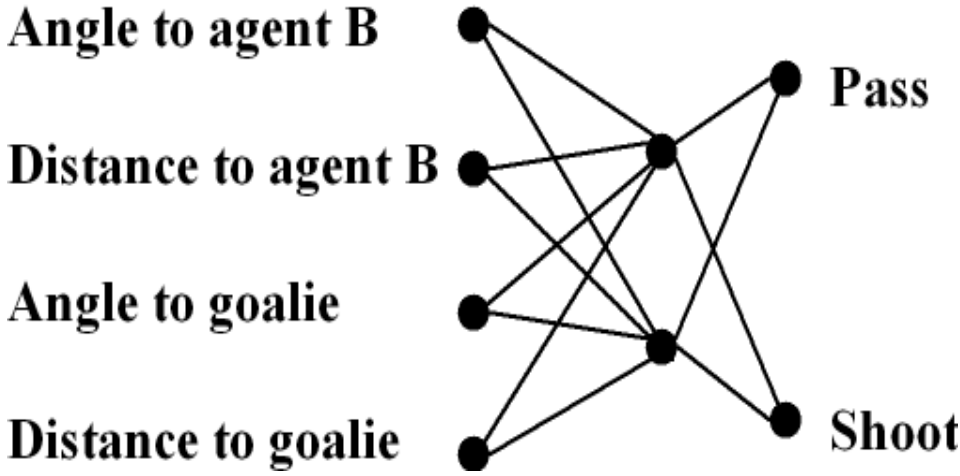


	Pass	Shoot	Total
Good	31.4%	52.1%	40.9%
Bad	68.6%	47.9%	59.1%

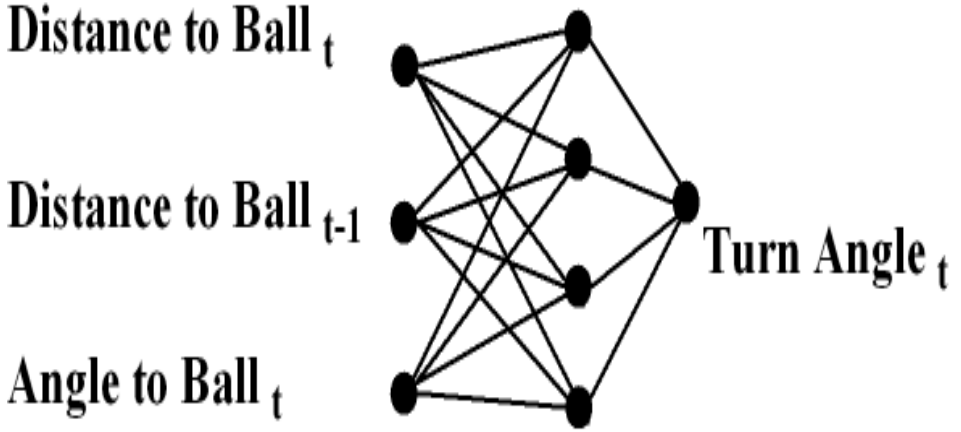

Direction to Teammate
Direction to Goal
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Output
-60
-50
-40
-30
-20
-10
0
10
20
30
Direction to Opponent Player
"pass.out"
"shoot.out"

Cooperative & Adversarial Behaviours

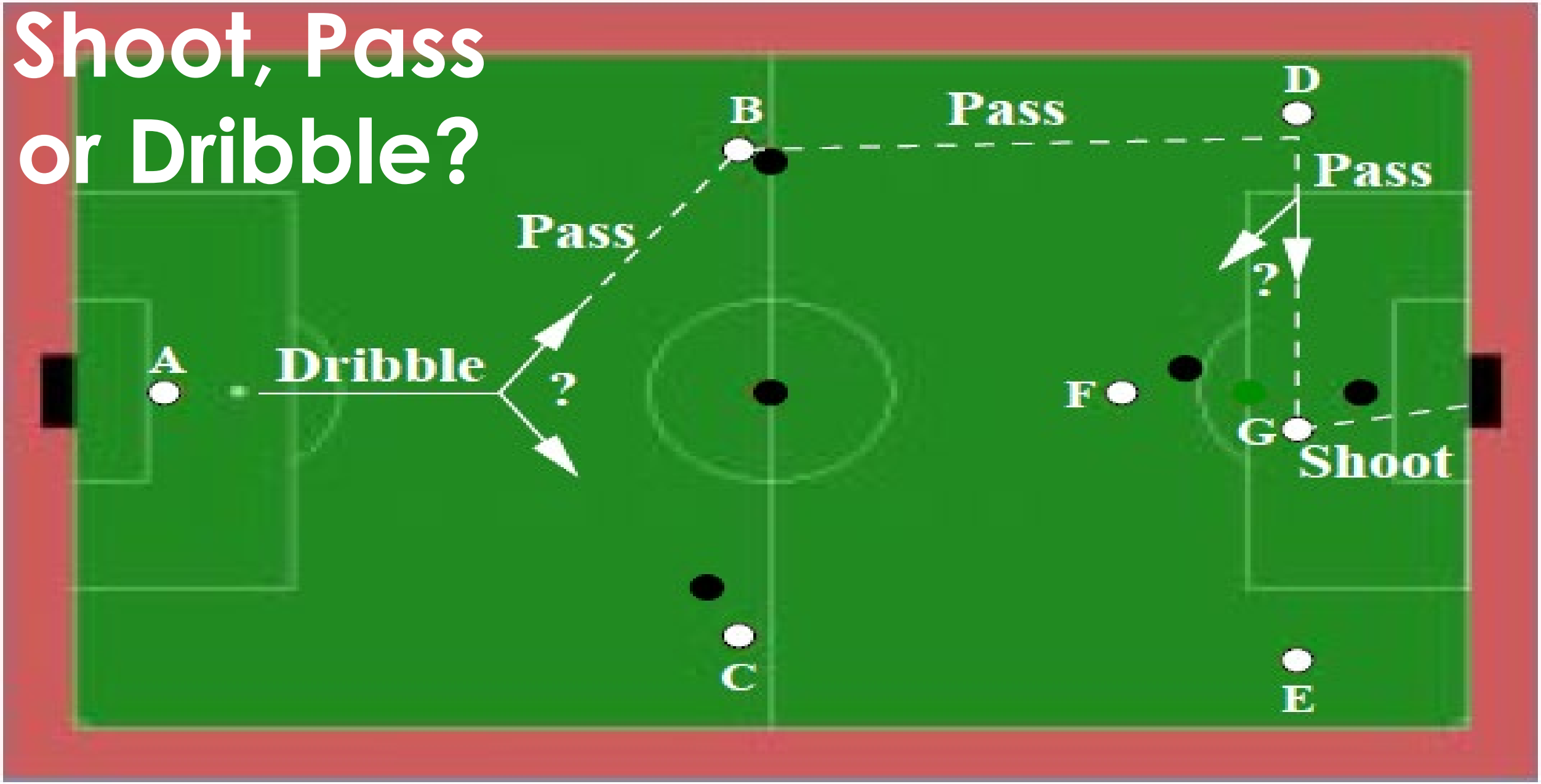
- **IF *Pass* OR *Shoot* (Agent A):**
  - "Pass-or-shoot" ANN (Trained 4-input ANN)
  - Ball is stationary.



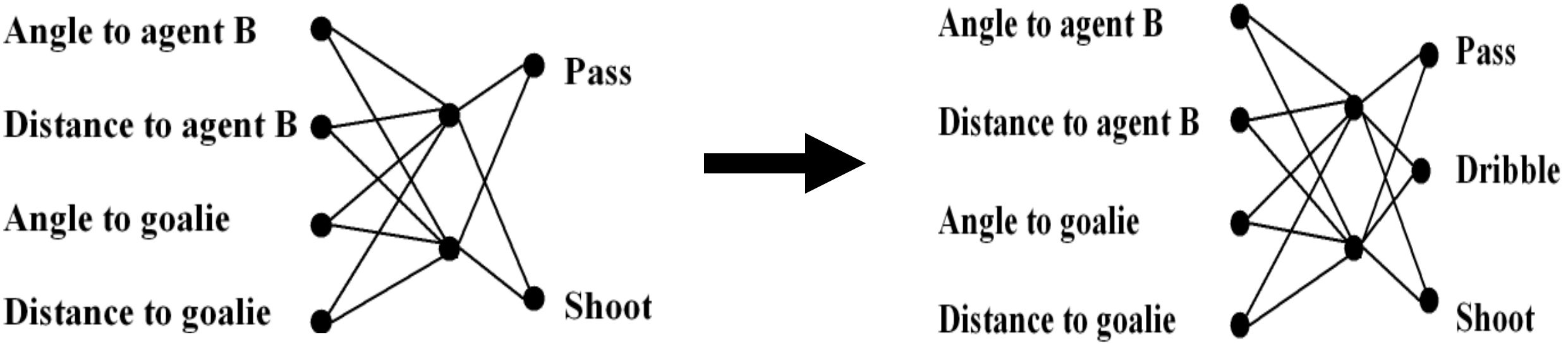
- **IF *Intercept* (Agent B, Goalie):**
  - Uses intercept ANN (Trained 3-input intercept ANN).
  - Ball is moving.



Shoot, Pass or Dribble?

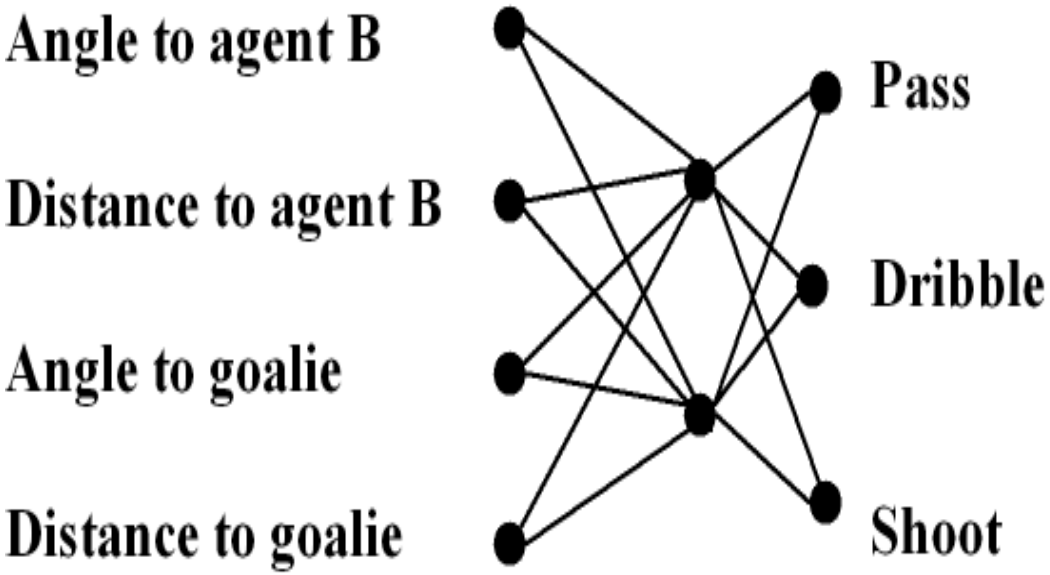


Agent A's ANN: Add Dribble Output



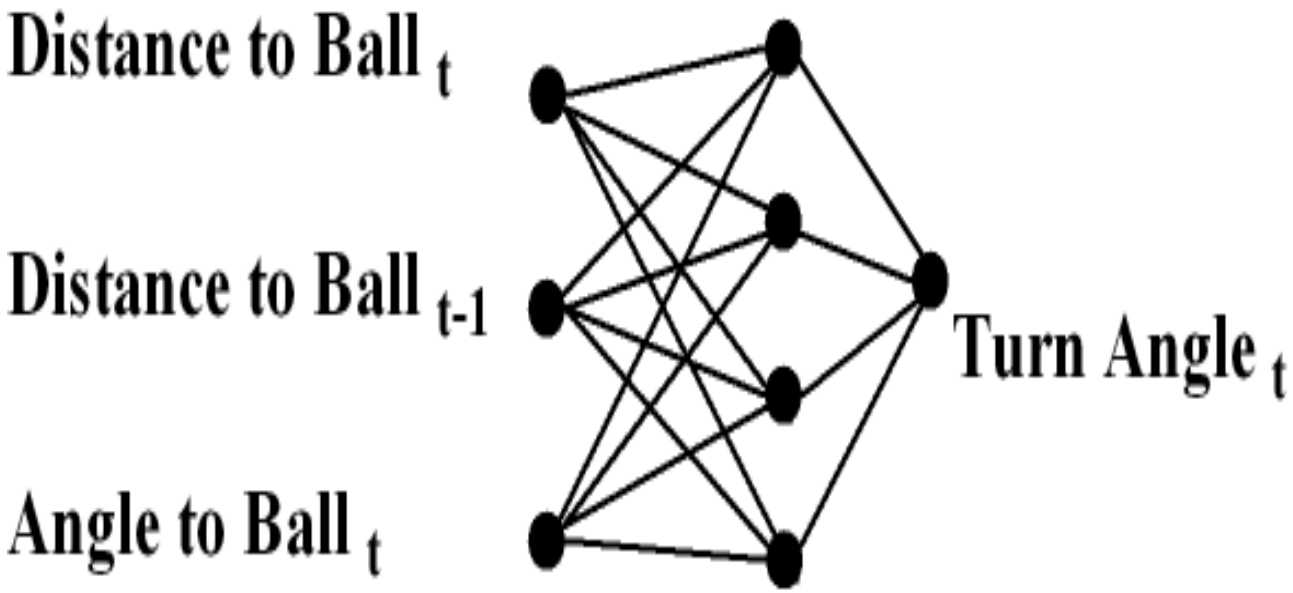
Pass, Shoot or Dribble ANN

- **IF** Ball is stationary or intercepted.
- **IF** Team-mates and opponents are nearby.
- **Output:** Pass, Dribble or Shoot.



Intercept Ball (Agent B, Goalie) ANN

- **IF** Ball moving towards my team's goal or towards me?
- **Output:** Angle to turn (Agent then moves at fixed speed).



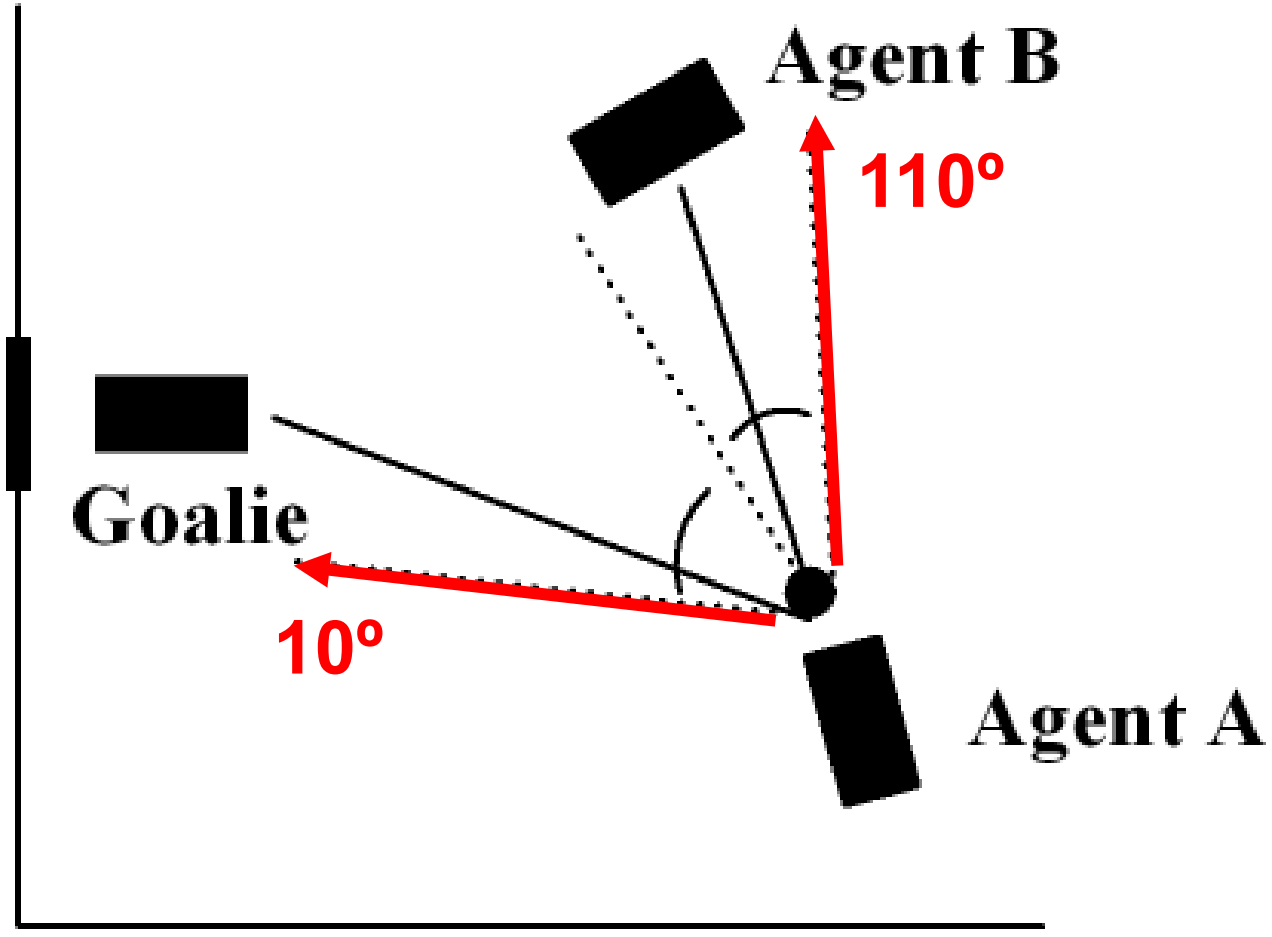
Collect More Training Data: Pass, Shoot, Dribble & Intercept ANNs

ANN1 output labels: [0, 1]:

{ *Shoot*, *Pass to B*, *Dribble* }

ANN2 output labels: [0.0, 90.0]:

{ *Intercept* }

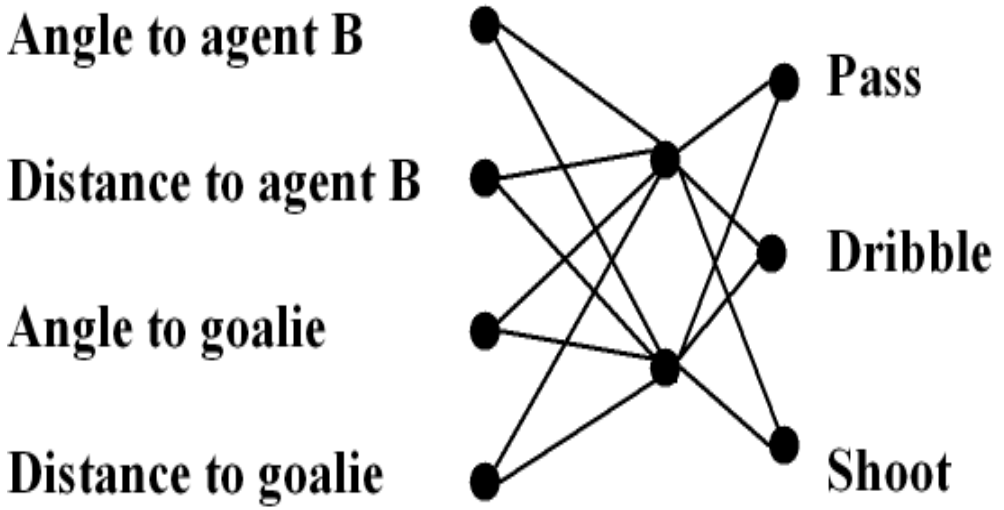


ANN1 Training example input:

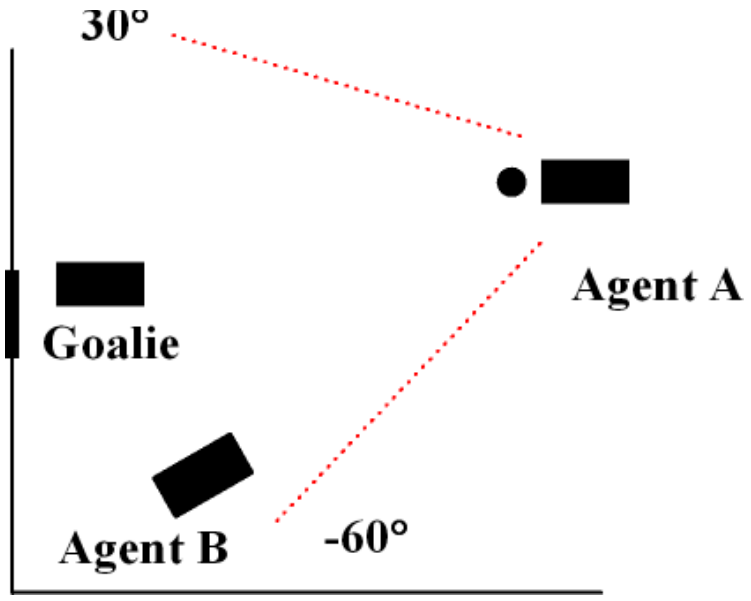
{ *Angle to B*, *Distance to B*, *Angle to Goalie*, *Distance to Goalie* }

Testing Agent A: Pass, Shoot, Dribble?

- **2000 Test simulations:**
  - Goalie, Agent B in random positions.
  - In angle range: **[-60º, 30º]** of **Agent A.**
  - Agent A's ANN receives 4 inputs: output?

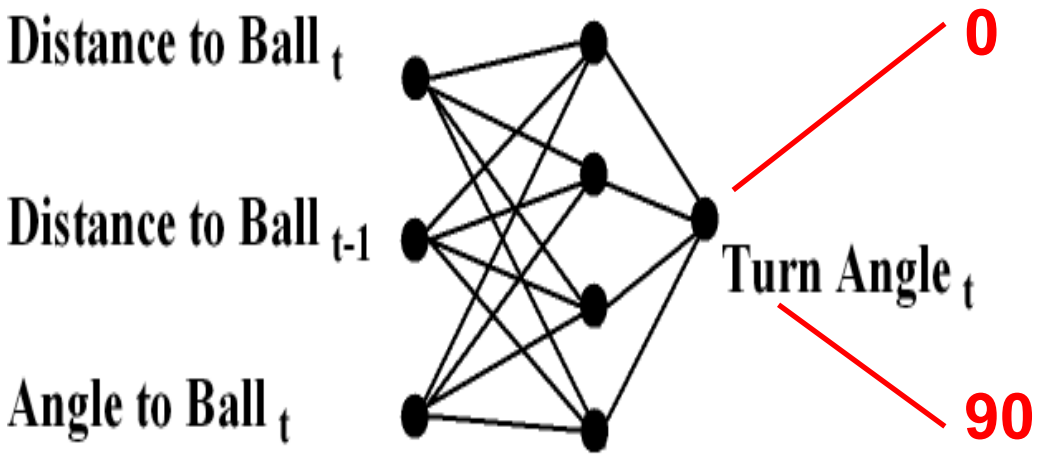


- **Highest activation → *Action selected* :**
  - Output Node 1: Pass.
  - Output Node 2: *Dribble*.
  - Output Node 3: *Shoot*.



Testing Agent B, Goalie: Intercept

- **2000 Test simulations (Agent B, Goalie):**
  - Agent B, Goalie ANN receives 3 inputs; Output: turn (move preset);

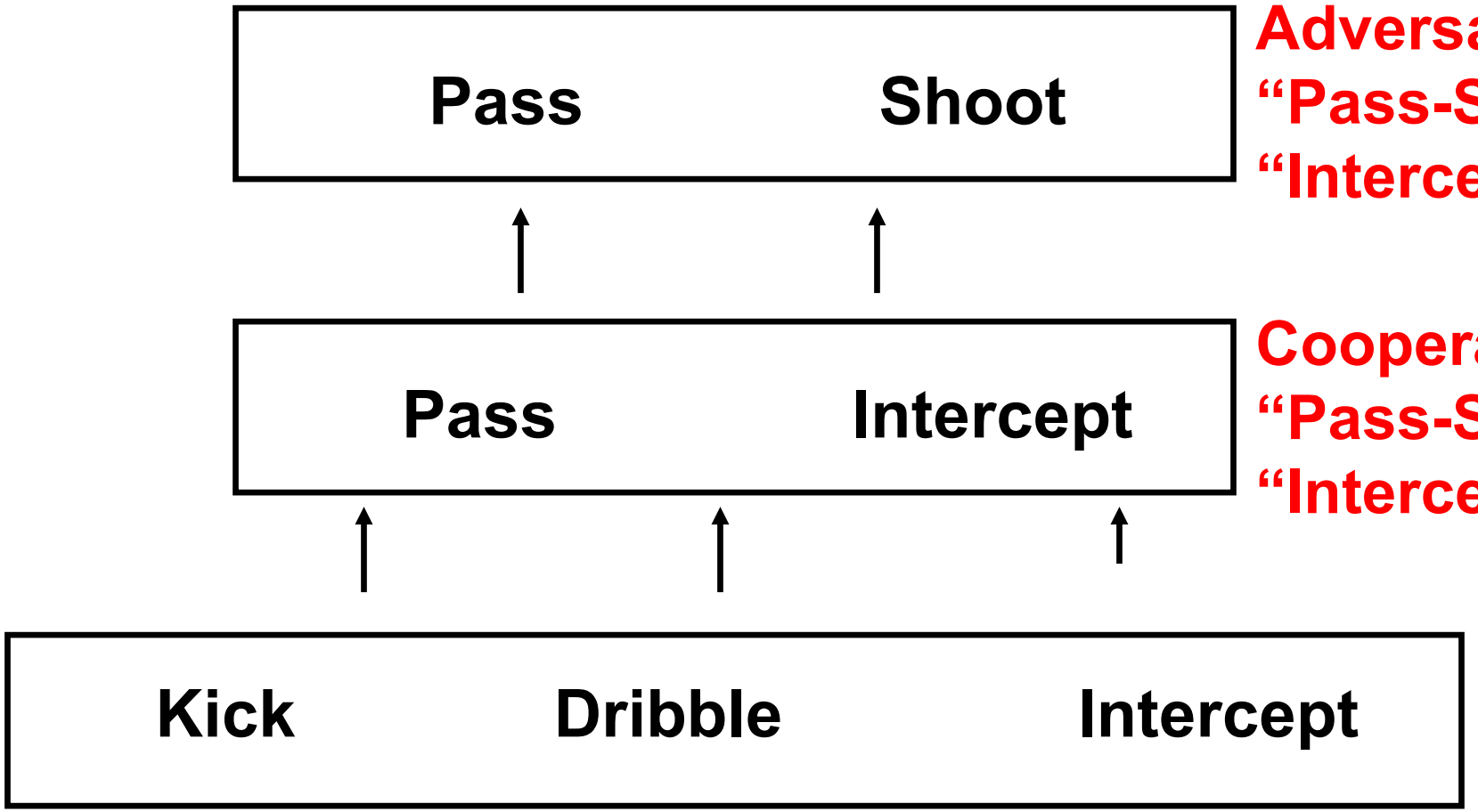


- **Node activation → *Action selected: Turn Angle [0, 90]***

How to get Team Behaviour?

- Agents use which-ever trained ANN suits the game state, e.g.:
  - If have ball, teammate and opponent in close proximity..
  - If opponent has ball, other opponent in close proximity..
  - If ball is moving towards *my team's* goal..

Putting all the Layers Together



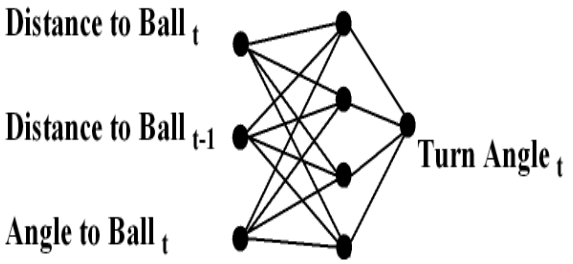
Adversarial (2 Teams: "Pass-Shoot-Dribble", "Intercept" ANNs)

Cooperative (1 Team: "Pass-Shoot-Dribble", "Intercept" ANNs)

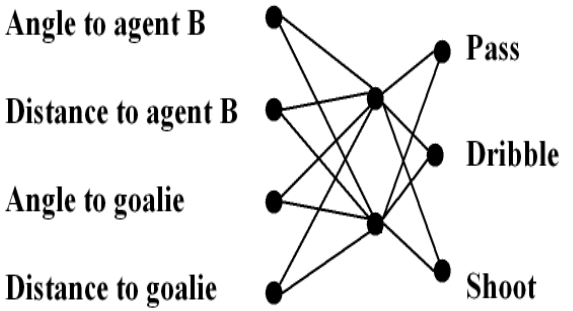
Individual (1 Agent: "Pass-Shoot-Dribble" ANN)

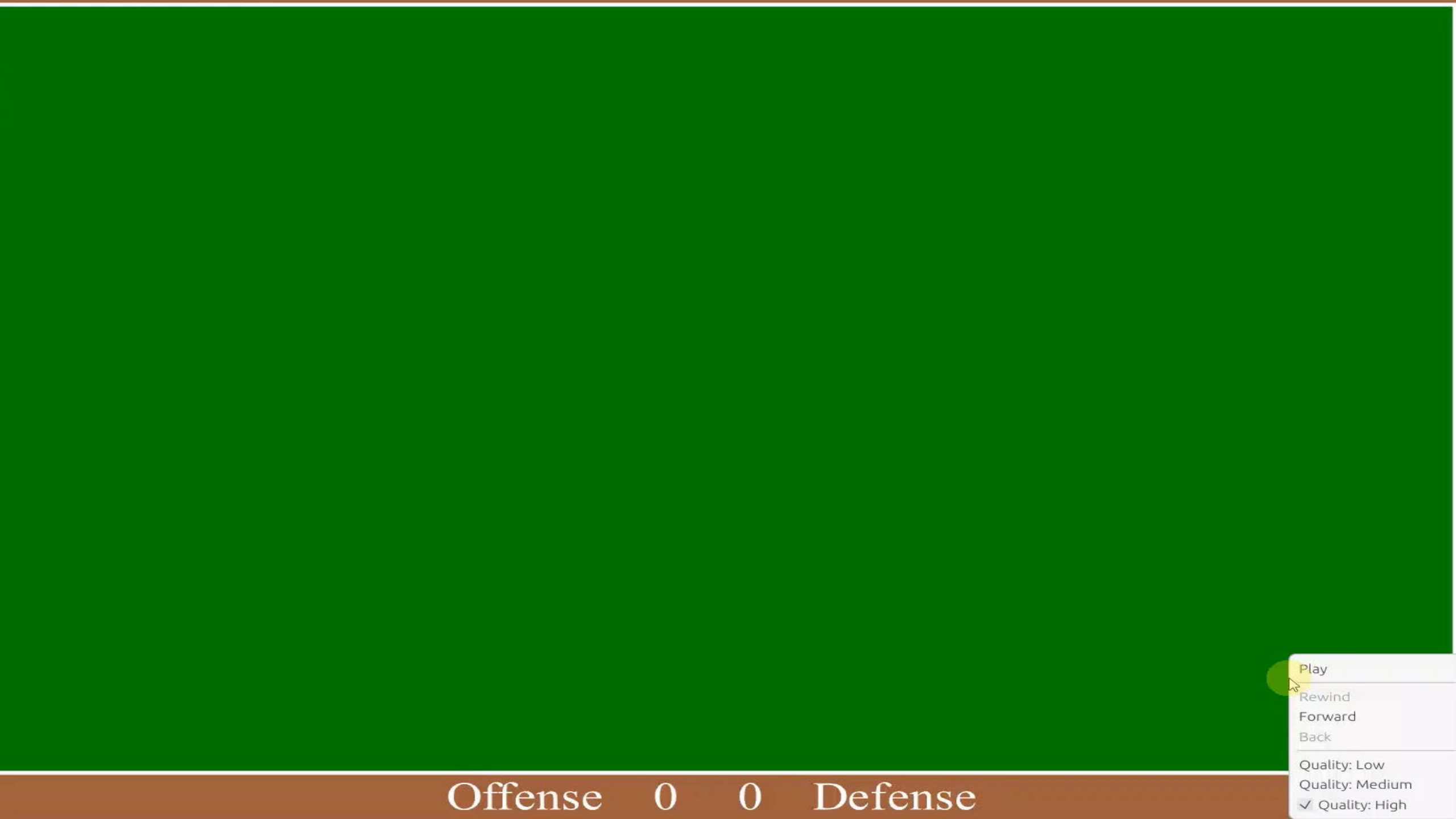
Random (Red) vs Trained (Blue) Team

- **Random Team: Each agent (A, B) uses fixed rules:**
  - **IF** no other team-mates closer, move straight to intercept ball;
  - **IF** have ball, randomly: *shoot*, *pass* or *dribble*.



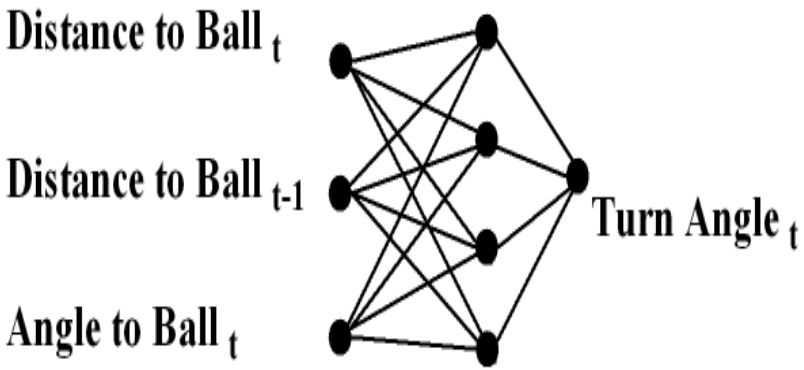
- **Trained Team: Each agent (A, B) uses either:**
  - Intercept (3-input) ANN.
  - Pass, shoot or dribble(4-input) ANN.



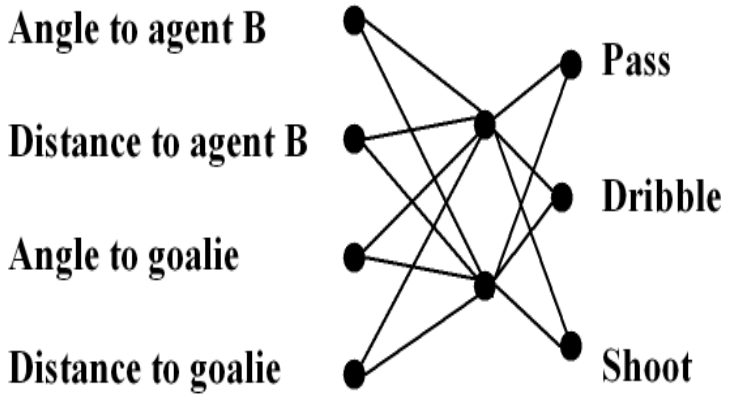

Offense 0 0 Defense
Play
Rewind
Forward
Back
Quality: Low
Quality: Medium
Quality: High

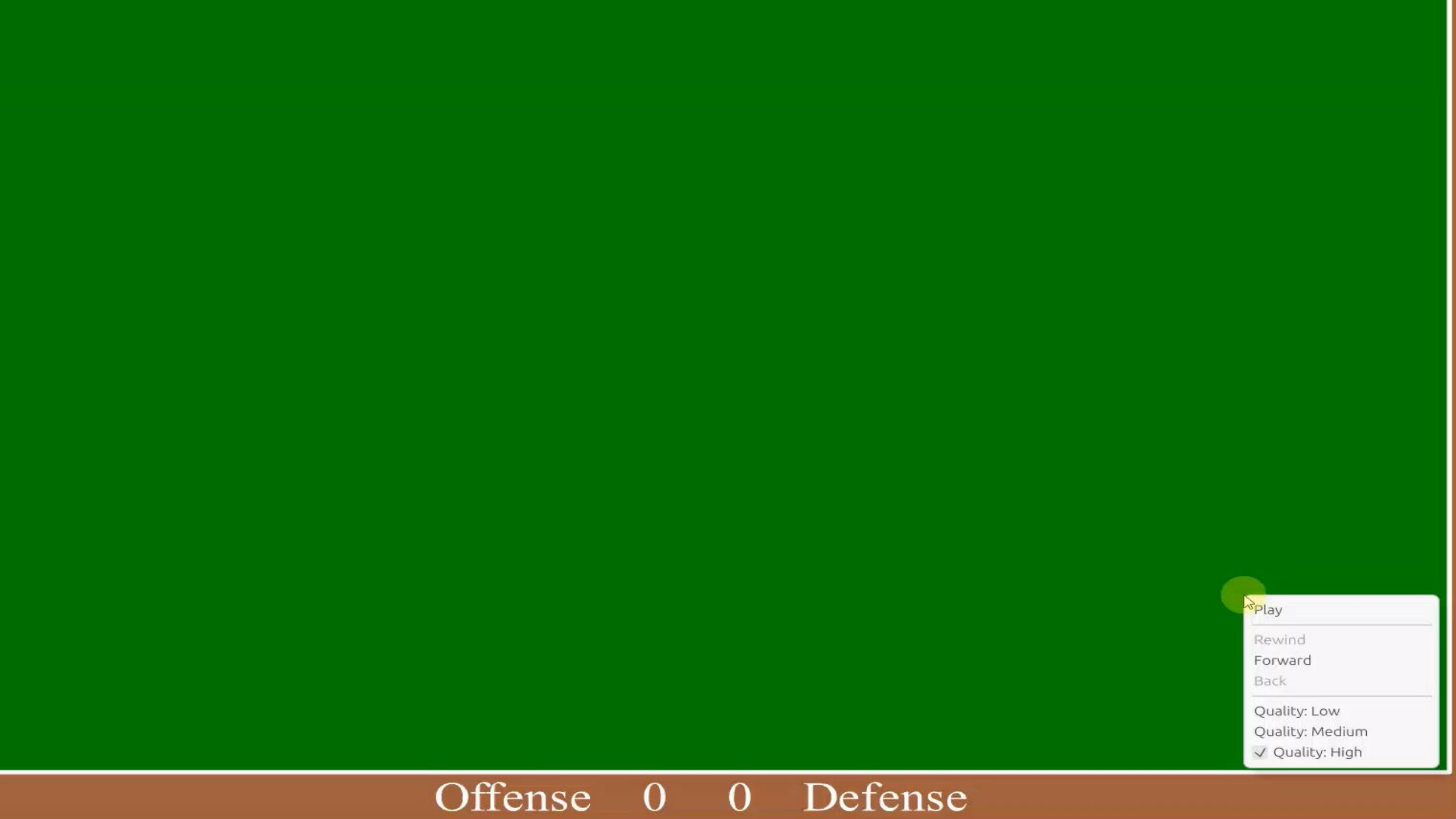
Trained (Red) vs Trained (Blue) Team

- **Trained Team: Each agent (A, B) uses either:**
  - Intercept (3-input) ANN:

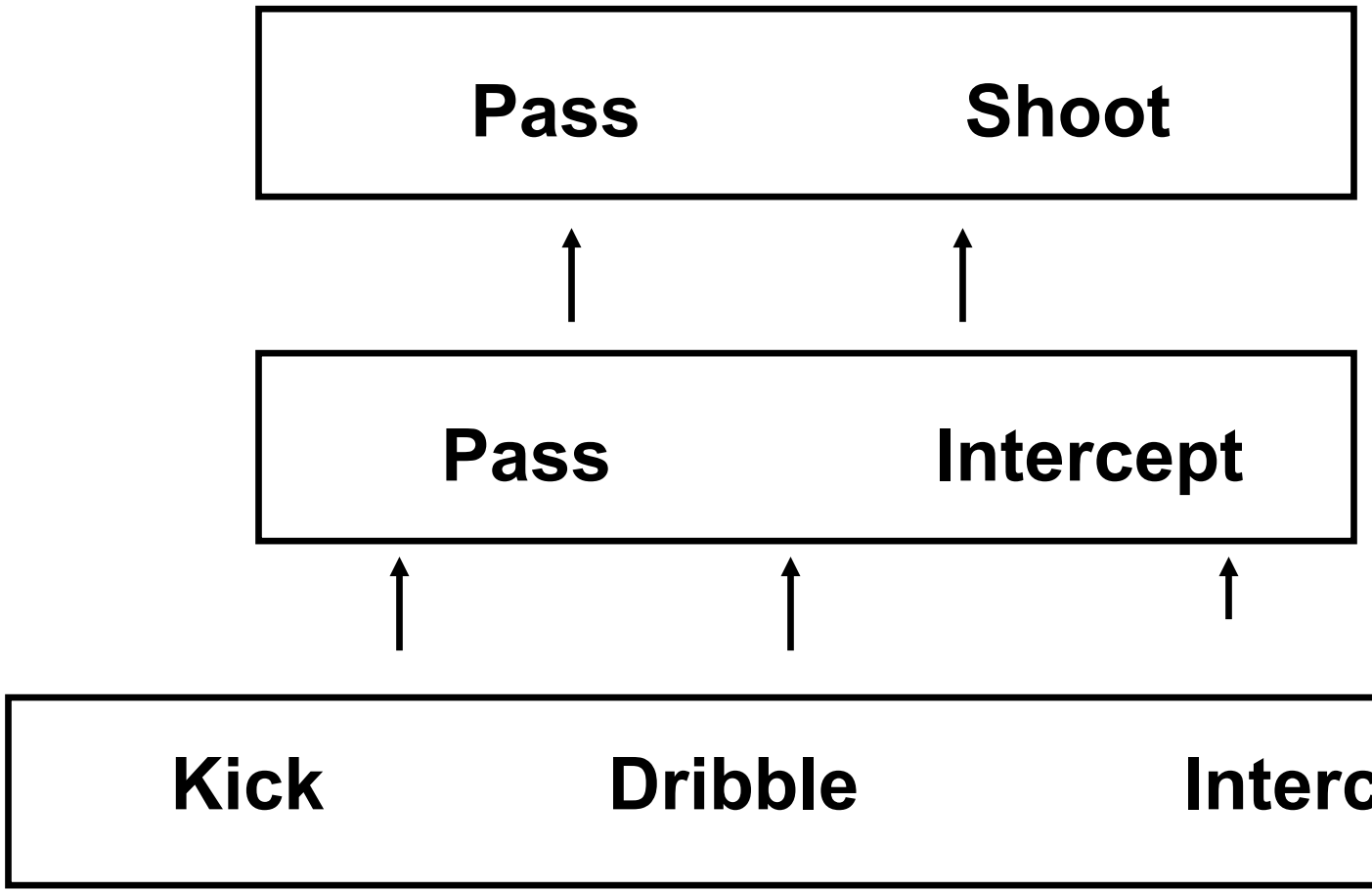


  - Pass, shoot or dribble (4-input) ANN:




Play
Rewind
Forward
Back
Quality: Low
Quality: Medium
Quality: High
Offense 0 0 Defense

Subsumption Architecture: Team Play



Adversarial (2 Teams: "Pass-Shoot-Dribble", "Intercept" ANNs)

Cooperative (1 Team: "Pass-Shoot-Dribble", "Intercept" ANNs)

Individual (1 Agent: "Pass-Shoot-Dribble" ANN)

Subsumption (Reactive) Architectures – Biggest Problem?

Reactive Architectures – Limitations

- **Lacks Memory, Learning:** Without memory, robots cannot learn from experience, store complex data, limiting long-term adaptability.
- **Limited Planning:** Architecture's reactive nature isn't suited for tasks requiring detailed planning or sequential steps.
- **Difficult to Adapt to New Tasks:** Each behaviour layer is task-specific, making it hard to modify for new objectives.
- **Designing Emergent Behaviour:** It is very difficult to design agents (behaviours) such that they interact to produce a (desired: problem-solving) emergent behaviour.

Summary: What you should know

- Reactive vs Deliberative vs Hybrid agent architectures.
- Pros and cons of each.
- How they work – Examples of each.

Next:
Belief, Desire, Intention (BDI) Agent Architecture

Reading (On Amathuba!)

- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* – 4th Edition – Ch2 (Intelligent Agents): aima.cs.berkeley.edu/
- Poole, D., & Mackworth, A. (2023) *Artificial Intelligence: Foundations of Computational Agents* – 3rd Edition – Ch1 (Artificial Intelligence and Agents): https://artint.info/3e/html/ArtInt3e.Ch1.html
- Krakauer, D., et al. (2026). Large language models and emergence: a complex systems perspective. Phil. Trans. R. Soc. A; 384 (2320): 20250014.
- Reynolds, C. (1987). Flocks, herds and schools: A distributed behavioral model. Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.
- Treuille, A., et al. (2006). Continuum crowds. ACM Transactions on Graphics, 25(3): 1160–1168.
- Theraulaz, G., & Bonabeau, E. (1995). Coordination in Distributed Building. Science 269: 686–688.

